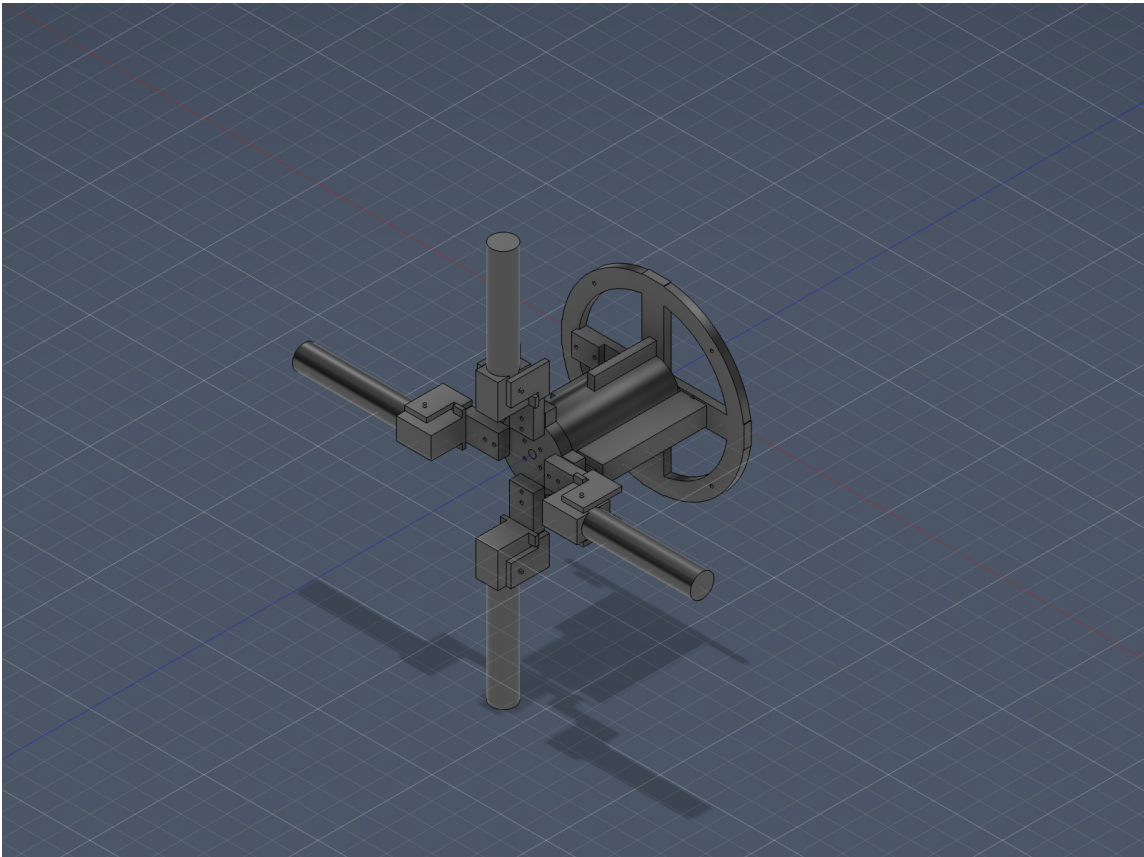


BLOOM-2 swinging-bucket centrifuge

Parametric Fusion 360 prototype for four 15 mL conical tubes and Ficoll/PBMC separation

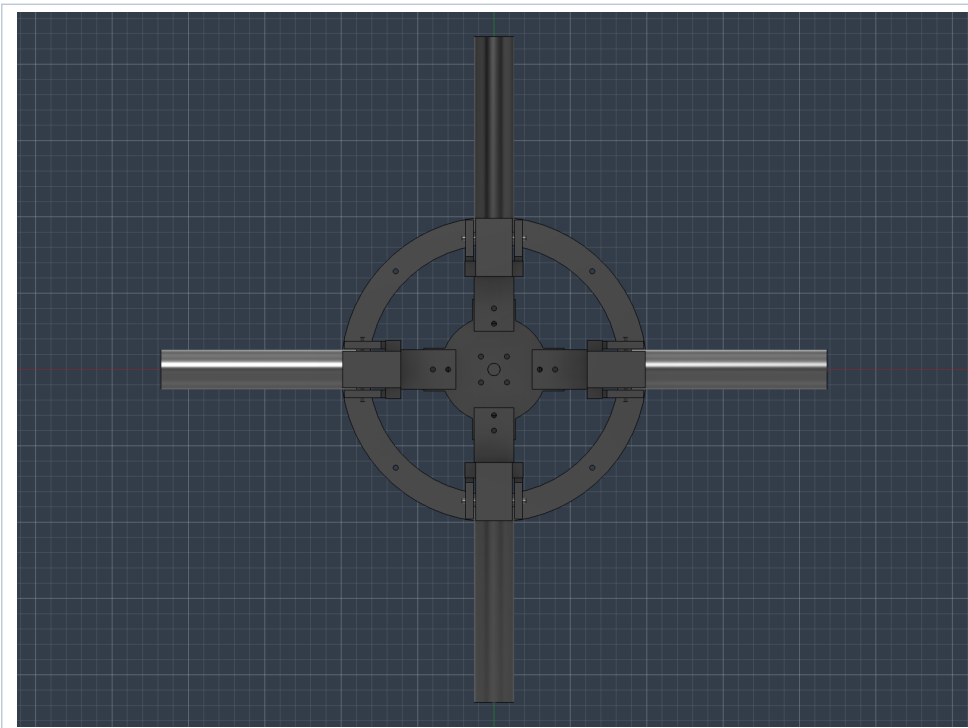


DESIGN POINT	1545 rpm
RESULTING RCF	400.3 x g
MID-FLUID RADIUS	150 mm
SWING PIVOT RADIUS	86 mm
AT-SPEED TIP RADIUS	218 mm
MINIMUM POT ID	466 mm
CAPACITY	4 x 15 mL
BUCKET TRAVEL	vertical to 90 deg

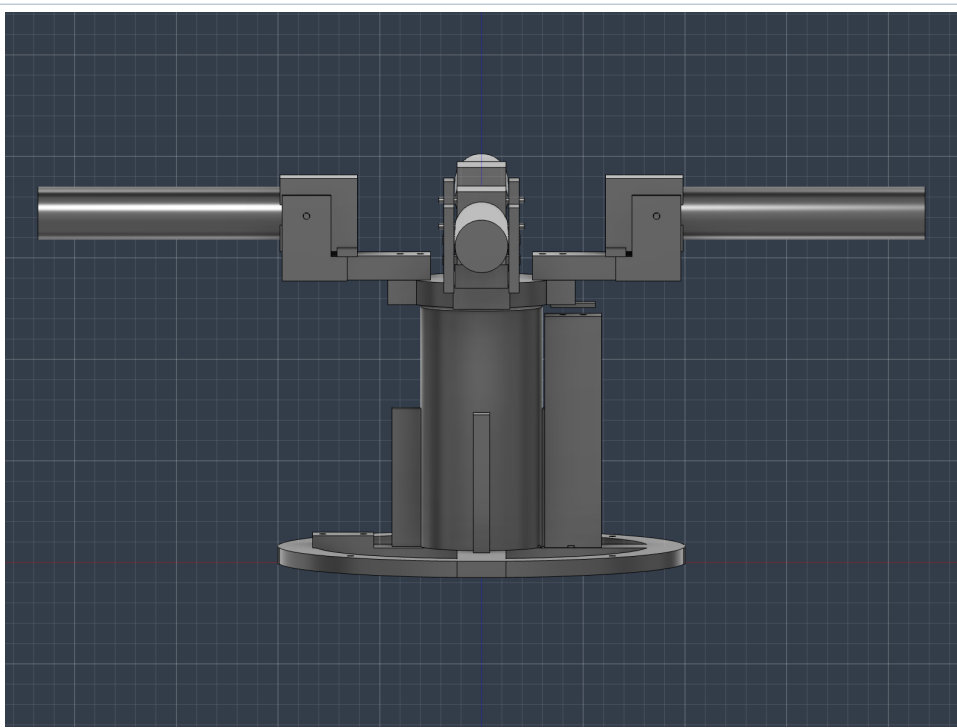
Configuration shown: four buckets at the 90-degree at-speed hard stop. Optional direct and geared motor plates are excluded from the main assembly views for clarity. The rotating assembly is split into one hub, four identical arms, and four identical buckets so every printed part fits a 220 x 220 mm bed.

Assembly views and interfaces

TOP - four-fold symmetry



FRONT - bearing stack and bucket stop



01	Dual 608ZZ tower	22.2 mm seats, 8.3 mm through bore; light axial preload and minimum shaft overhang.
02	Shared rotor interface	Two M3 fasteners per arm. Four-hole 24 mm BCD accepts direct coupler or driven gear without reprinting the rotor.
03	Free swinging bucket	M3x45 tangential pivot, 0.5 mm clevis clearance, low-friction washers, and printed 90-degree stop.
04	Instrumentation	Tall IR boss looks upward to the underside rotor target; MPU-6050 bolts rigidly to the chassis boss.
05	Containment envelope	436 mm diameter rotating envelope plus 15 mm wall clearance on both sides gives a 466 mm minimum clear pot ID.

Bill of materials and manufacturing notes

Category	Item	Qty	Specification	Notes
Printed	Chassis and bearing tower	1	PETG high infill	Includes IR pedestal and rigid MPU boss
Printed	Rotor hub	1	PETG solid at bolt zones	Four shared arm lands and drivetrain interface
Printed	Rotor arm	4	PETG high infill	Print together and mass match
Printed	Swing bucket	4	PETG high infill	18.6 mm bore for 17 x 120 mm tube; print together
Printed	Direct motor plate	1 optional	PETG	Use only for smooth direct-drive startup
Printed	Direct coupler	1 prototype	PETG prototype	Commercial 3.17-to-8 mm coupler preferred
Printed	Gear motor plate	1 optional	PETG	36 mm nominal center distance
Printed	Driven gear	1 optional	60T module-1 prototype	Replace simplified teeth with validated involute
Printed	Motor pinion	1 optional	12T module-1 prototype	Replace simplified teeth with validated involute
Owned	A2212 brushless motor	1	1000 KV	Ticket-specified
Owned	Sensorless ESC	1	30 A	Disable brake and program smooth ramp
Owned	Battery	1	2S LiPo approximately 7.4 V	Add fuse and physical disconnect
Owned	Bearing	2	608ZZ 8 x 22 x 7 mm	Light preload only
Hardware	Shaft	1	8 mm diameter x approximately 100-125 mm	Ground steel preferred
Hardware	Bucket pivot screw	4	M3 x 45 mm	Use smooth shank where possible
Hardware	Bucket pivot nylock nut	4	M3	Do not pinch clevis
Hardware	Low-friction pivot washer	8	M3 PTFE or acetal	One each side of every bucket lug
Hardware	Arm-to-hub screw	8	M3 x 20 mm	With washers and nylock nuts
Hardware	Drive-interface screw	4	M3 x 16 mm	Shared direct/gear bolt pattern
Hardware	Motor/chassis screw set	1	M3 assorted	Fit selected plate and A2212 cross mount
Instrumentation	IR reflective sensor	1	Reflective optical tachometer	Align to rotor underside target
Instrumentation	MPU-6050 module	1	Rigid chassis mount	No vibration isolation
Consumable	15 mL conical tube	4	17 x 120 mm	Opposing loaded tubes matched to +/-0.1 g
Containment	Steel pot and lid	1	Minimum 466 mm clear inside diameter	Lid closed for every powered test

PRINT	PETG only. Print hub and arms flat. Print buckets on their side so centrifugal load runs in the layer plane. Use 6 perimeters, 8 top/bottom layers, and 70-100% infill with solid pivot/bolt zones. Print each matched set in one job.
BALANCE	Match printed buckets to +/-0.5 g. Match opposing loaded tubes to +/-0.1 g before every run. All four empty buckets must fall freely to the same hard stop.
CONTROL	Smooth spin-up, no torque steps, and coast-only stop longer than 2 minutes. Disable active brake. Use the 5:1 option only if direct drive cannot start and hold the loaded rotor smoothly.
VALIDATE	Prototype only. Containment and guarded overspeed testing precede biological use. Final functional test: four mass-matched tubes, dyed water over glycerol, 1545 rpm for 30 min, coast down. Clean interface passes; smear fails.